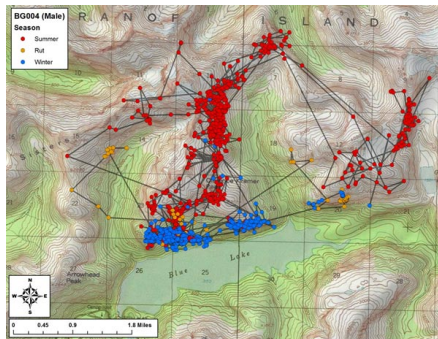




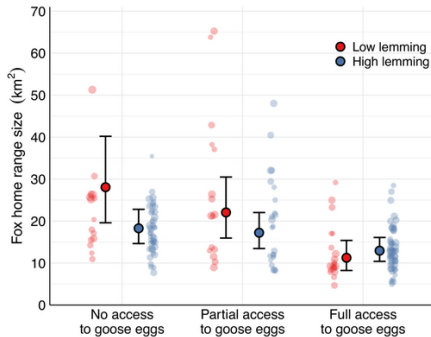
From individuals to collective behaviour

Dominique Gravel



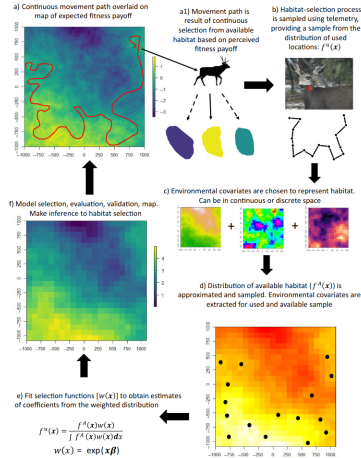


Statistical analysis





Resource-selection function





Outline

- ▶ Random walk and diffusion
- ▶ Mechanistic home range analysis
- ▶ Adding population dynamics
- ▶ Predator-prey interactions
- ▶ Self-organization & complex spatial structures

Exercise



A minimal individual-based model

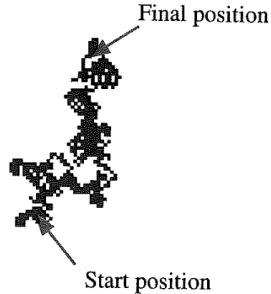
- ▶ Start at position (0,0)
- ▶ At each time step add $\Delta x = \eta(0, \sigma)$ and $\Delta y = \eta(0, \sigma)$
- ▶ Record position at time $t = 1, 2, 4, 6, 8, 16, 32, 64$ time steps
- ▶ Repeat 100 times
- ▶ Plot the mean and variance of distance to the center as a function of time

Theory



Random walk

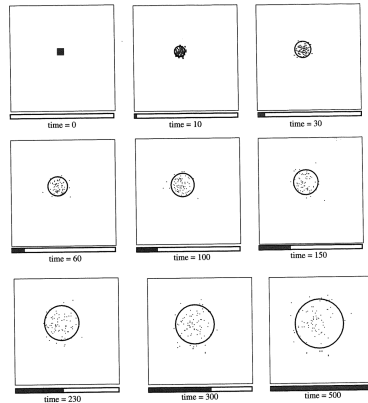
Definition : a stochastic process that describes a path that consists of a succession of random steps on some mathematical space





Diffusion

Definition : the population of consequence of many individuals simultaneously undergoing random walks in continuous time and across continuous space





Numerical example

- ▶ Start with $N = 2000$
- ▶ Each time step, $1/10$ individuals ($d = 0.1$) move to the left and the same to the right

	x=1	x=2	x=3	x=4	x=5	x=6
Time 0			1000	1000		
Time 1		100	900	900	100	
Time 2	10	170	820	820	170	10



More generally

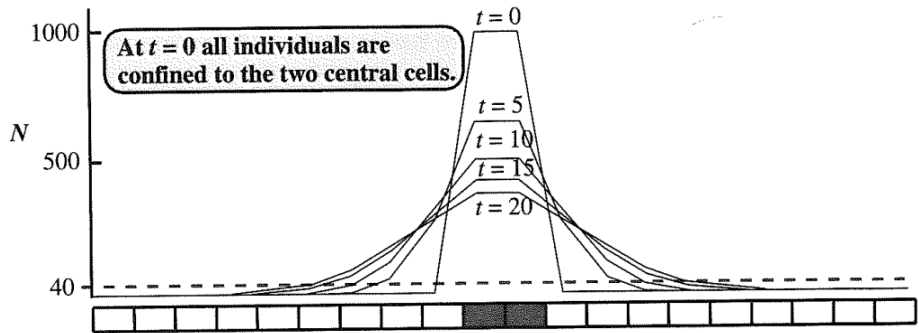
Density change per cell follows :

$$N_{x,t+1} = N_{x,t} + dN_{x-1,t} + dN_{x+1,t} - 2dN_{x,t}$$



Result

Which looks like :





From discrete to continuous

We can reformulate the equation as a change in density at location x :

$$\Delta N_x = N_{x,t+1} - N_{x,t} = d[(N_{x-1,t} - N_{x,t}) + N_{x+1,t} - N_{x,t}]$$

As the time and spatial steps get smaller, the formula becomes :

$$\frac{dN(x,t)}{dt} = D \frac{\partial^2 N(x,t)}{\partial^2 x}$$

Where :

- ▶ The left term is the rate at which the density at position x is changing over time
- ▶ D is the diffusion parameter
- ▶ The right term is a density gradient at location x



Diffusion coefficient

Units are *distance*² per time

Interpretation is the mean square displacement (one dimension):

$$D = \frac{\overline{\delta^2}}{2t}$$



Solution

The number of individuals released at a location will follow a normal distribution :

$$N(x, t) = \frac{N_0}{\sqrt{4\pi Dt}} \exp \left(\frac{-(x - \bar{x})^2}{4Dt} \right)$$

With interpretation :

- ▶ The mean displacement expands across space at a rate proportional to the square root of time
- ▶ The total area (range size) increases proportionally with time



Adding advection

$$\frac{dN(x,t)}{dt} = \underbrace{-\nabla CN(x,t)}_{\text{advection}} + \underbrace{\nabla^2 DN(x,t)}_{\text{diffusion}}$$

Where ∇ is a short notation for the partial derivative of position over space and c is the advection rate

Exercise



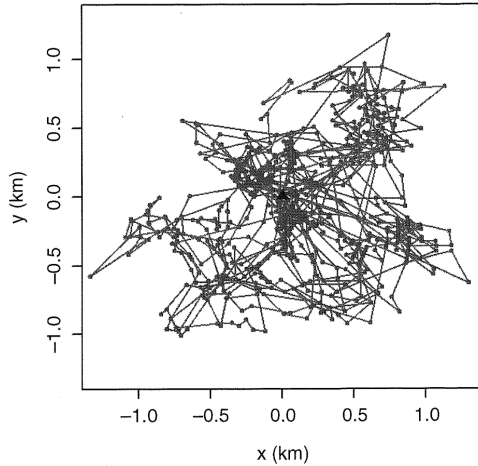
A minimal individual-based model Adding a den site

What happens if movement changes as the animal gets further away from the den ?

Back to theory

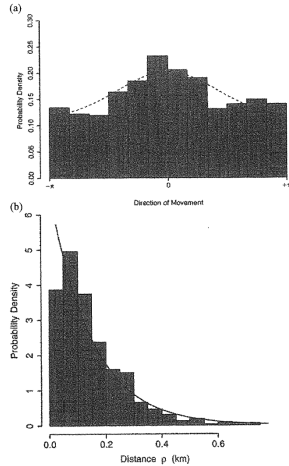


Red fox telemetry





Mechanistic home range analysis





Redistribution kernel

The probability density function describing movement from location x to x' :

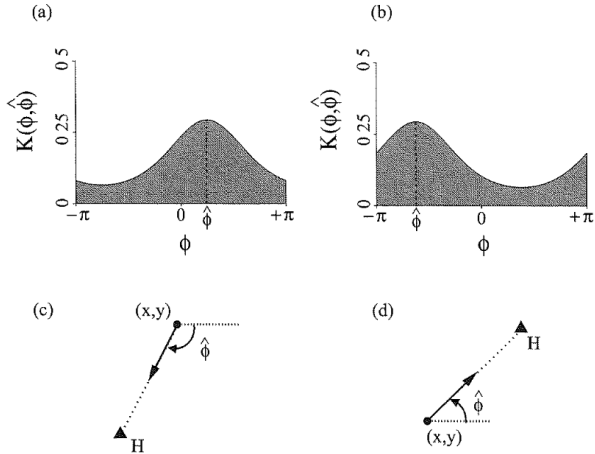
$$k(x, x', \tau, t) = \frac{1}{\rho}(\rho) K_{\tau}(\phi, \Theta)$$

where

- ▶ ρ is the distance between locations
- ▶ ϕ is the angle between starting point and finishing point
- ▶ Θ is the direction of the home range center from the current position



Redistribution kernel





Equation for space use

We can revisit the equation for advection-diffusion and compute parameters from the redistribution kernel (hidding complications with 2 dimensions):

$$\frac{dN(x,t)}{dt} = \underbrace{-\nabla CN(x,t)}_{\text{advection}} + \underbrace{\nabla^2 DN(x,t)}_{\text{diffusion}}$$

where :

- ▶ $C = \kappa_\tau \bar{\rho}_\tau / 2\tau$
- ▶ $D = \bar{\rho}_\tau^2 / 4\tau$

At equilibrium : diffusion equals advection (movement toward the center of the home range)



Solving for home range

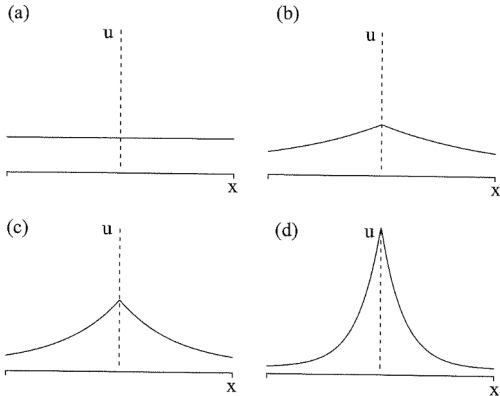
After a bit of (tedious) maths, we get a fairly simple solution :

$$N(x) = \frac{\beta}{2} \exp(-\beta|x|)$$

where $\beta = C/D$



Solving for home range



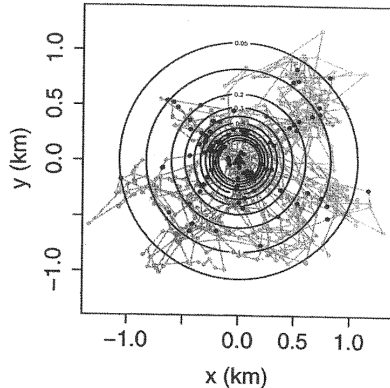
For a) $\beta = 0$, b) $\beta = 1$, c) $\beta = 2$ and d) $\beta = 5$.

Note that the same results hold quantitatively for 2 dimensions.



Solving for home range

We now have a probability density function emerging from first principles and basic statistical descriptors of movement :





Movement plus geometric growth

Recall the original discrete time discrete space equation :

$$N_{x,t+1} = N_{x,t} + dN_{x-1,t} + dN_{x+1,t} - 2dN_{x,t}$$

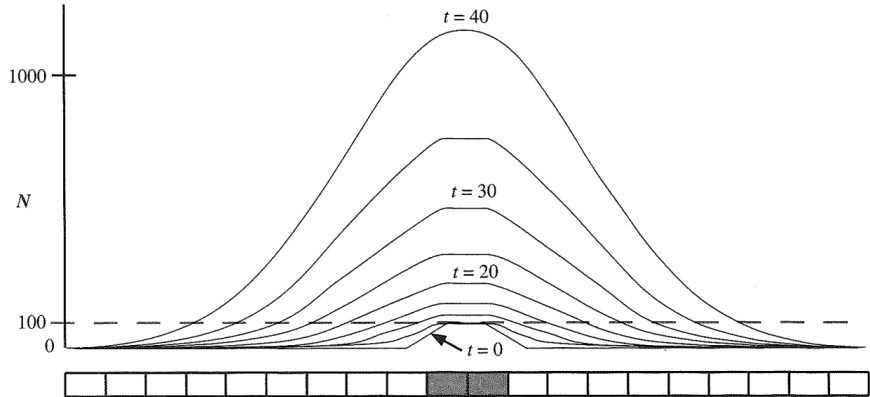
And now add geometric growth :

$$N_{x,t+1} = rN_{x,t} + dN_{x-1,t} + dN_{x+1,t} - 2dN_{x,t}$$

What's your intuition ?



Movement plus geometric growth





Movement plus geometric growth

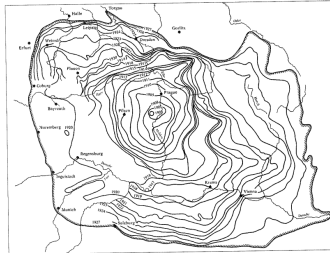
Solution is

$$Radius^2(t) = 4Dt[rt + \ln N_0]$$

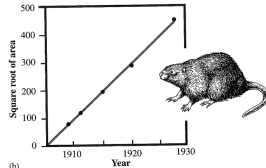
Which means :

- ▶ Range size (m^2) increases proportional to time
- ▶ Range limit (m) moves proportional to $\sqrt{r}$ and $\sqrt{D}$

Invasive muskrats Skellam, 1951



(a)



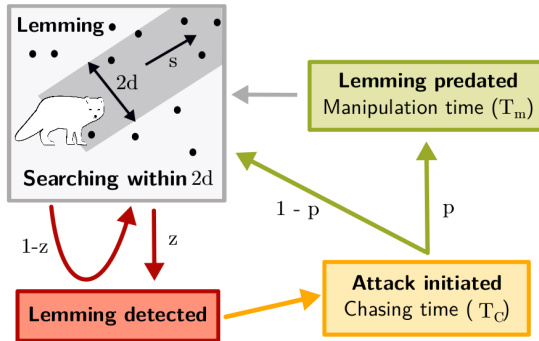
(b)





A mechanistic approach

C) Lemmings



Beardsell et al. 2022. Front. Ecology



Law of mass action

Principle : predator and prey are randomly moving particles, an interaction occurs whenever a predator encounters a prey, detect it and attack it with success

The *capture efficiency* ($km^2/time$) of predator individual (or species) i is :

$$\alpha_i = s_i z_i k_i p_i$$

Where

- ▶ s_i is average movement rate ($km/time$)
- ▶ z_i is reaction radius (km)
- ▶ k_i is detection probability
- ▶ p_i is attack probability

Note : we need to consider both the movement rate of the predator and the prey when both are moving in space



Available time for search

Principle : predators need time to catch, consume and process prey. This time is not available for foraging, thereby reducing search efficiency.

The *handling time* is the time required to catch and process a prey. It is split into two components :

- ▶ Time spent searching $\frac{T_{ci}}{p_i}$
- ▶ Time spent manipulating prey T_{mi}

Which yields :

$$h_i = \frac{T_{ci}}{p_i} + T_{mi}$$



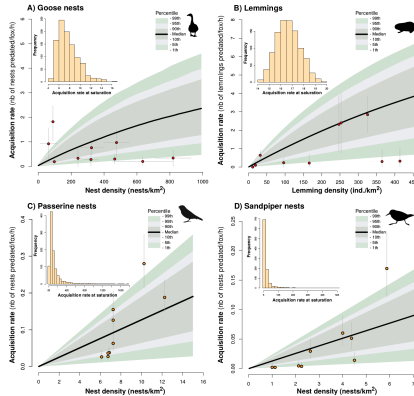
Integrating the whole thing

Putting pieces together we get the famous Holling Type II functional response that we will challenge this week :

$$f(i) = \frac{\alpha_i N_j}{1 + \alpha_i h_i N_i}$$



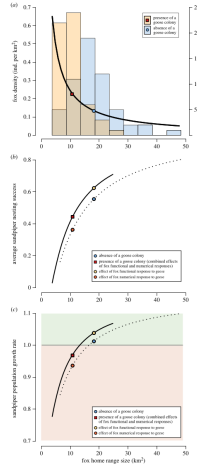
Application for the Arctic fox



Beardsell et al. 2022. Front. Ecology



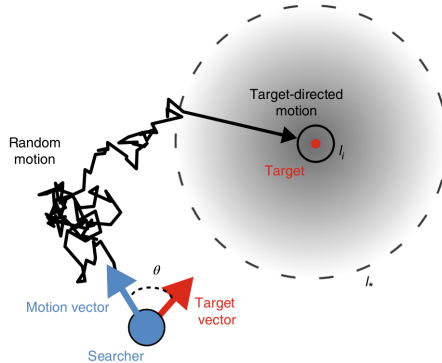
Non-stationary movement



Beardsell et al. 2023. Proc. Roy. Soc. London B.

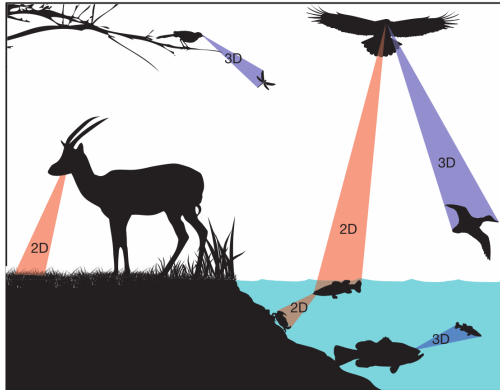


Non-random movement



Hein et al. 2022. Nat. Eco. Evo.

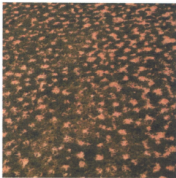
Dimensionality



Pawar et al. 2012. Nature



Patch formation in arid ecosystems



Rietkerk et al. 2002. Am. Nat

$$\frac{\partial P}{\partial t} = c \times g_{\max} \times \frac{W}{W + k_1} \times P - d \times P + D_p \Delta P,$$

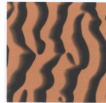
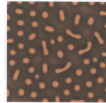
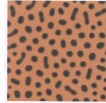
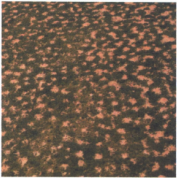
$$\frac{\partial W}{\partial t} = \alpha \times O \frac{P + k_2 \times W_0}{P + k_2} - g_{\max} \times \frac{W}{W + k_1}$$

$$\times P - r_w \times W + D_w \Delta W,$$

$$\frac{\partial O}{\partial t} = R - \alpha \times O \frac{P + k_2 \times W_0}{P + k_2} + D_o \Delta O,$$



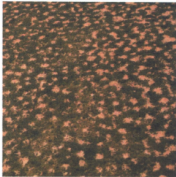
Patch formation in arid ecosystems



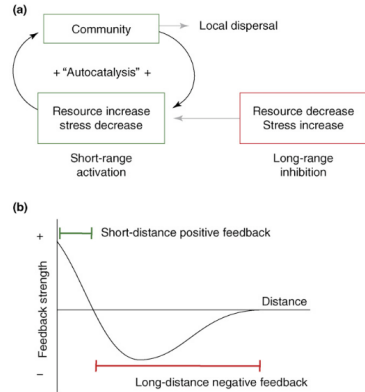
Rietkerk et al. 2002. Am. Nat



Turing instabilities

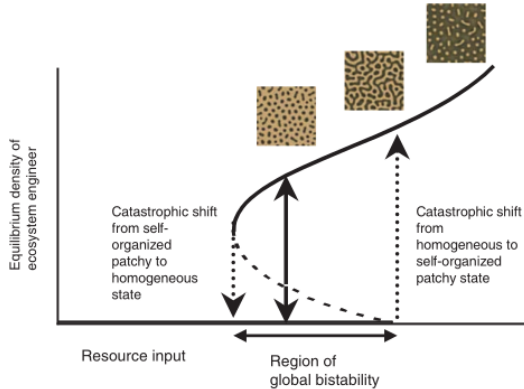


Rietkerk et al. 2002. Am. Nat





Patch formation in arid ecosystems



Rietkerk et al. 2004. Science

Exercise



Range dynamics under global changes

Discuss in group how biotic interactions can affect species range shift under climate change